

LARNE, United Kingdom

WMO: 039280

Lat: 54.852N

Lon: 5.831W

Elev: 10

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 86-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.7	34.5	26.4	20.4	36.2	28.4	22.4	37.4	46.1	41.7	39.5	43.0	8.8	240	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	69.1	62.2	66.9	60.8	64.8	59.6	63.6	67.7	61.8	65.6	60.2	63.7	10.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	82.3	66.2	59.9	77.0	64.3	58.5	73.2	62.6	28.9	68.1	27.7	65.7	26.6	64.0	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.9	30.1	28.3	DB	30.3	74.5	1.9	3.3	28.9	76.9	27.8	78.8	26.7	80.7	25.3	83.1	(4)
(5)			WB	28.4	67.2	2.9	3.2	26.3	69.5	24.6	71.3	23.0	73.1	20.9	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	42.6	42.3	44.8	46.5	51.5	55.4	58.7	58.7	55.4	51.6	47.3	44.0	(6)
(7)		DBStd	7.01	4.08	4.20	4.07	3.77	3.85	3.48	3.15	3.38	3.24	4.42	3.92	4.74	(7)
(8)		HDD50	1092	231	217	167	119	27	1	0	0	1	36	98	195	(8)
(9)		HDD65	5502	694	635	625	556	418	289	199	199	287	416	532	652	(9)
(10)		CDD50	1071	2	2	7	13	74	163	268	269	164	85	16	8	(10)
(11)		CDD65	4	0	0	0	0	0	0	2	2	0	0	0	0	(11)
(12)		CDH74	8	0	0	0	0	0	0	6	2	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	12.3	15.5	15.1	13.5	12.5	10.9	9.5	8.9	9.7	10.9	13.1	13.8	14.1	(14)
(15)	Precipitation	PrecAvg	44.0	4.6	3.5	3.1	3.0	3.0	3.1	3.1	3.3	3.3	4.8	4.5	4.6	(15)
(16)		PrecMax	56.5	7.9	7.4	6.9	5.3	8.5	7.3	7.2	9.1	7.6	8.9	8.9	9.5	(16)
(17)		PrecMin	35.0	1.2	0.9	0.6	0.7	0.4	1.1	1.0	0.4	0.0	1.5	2.3	2.3	(17)
(18)		PrecStd	5.3	1.6	1.7	1.4	1.2	1.7	1.7	1.3	1.9	1.9	2.1	1.7	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	54.9	57.5	61.2	65.9	70.3	73.4	72.2	67.2	63.6	58.4	56.9	(19)
(20)			MCWB	52.1	50.8	52.1	55.9	58.5	60.9	64.5	65.5	60.4	58.8	56.0	54.5	(20)
(21)		2%	DB	52.6	52.8	54.8	57.4	62.3	66.1	69.2	68.9	64.4	60.9	56.2	54.8	(21)
(22)			MCWB	50.0	49.4	50.4	53.5	56.1	59.0	62.3	62.8	59.4	57.6	53.6	52.5	(22)
(23)		5%	DB	50.9	51.0	53.0	54.5	59.7	63.3	66.8	66.6	62.2	59.4	54.6	52.9	(23)
(24)			MCWB	48.4	47.8	49.3	50.5	54.4	57.6	60.7	61.4	57.9	56.3	52.1	50.6	(24)
(25)		10%	DB	49.4	49.1	51.2	52.5	57.6	61.2	64.4	64.6	60.8	58.0	53.2	51.2	(25)
(26)			MCWB	47.2	45.9	47.9	48.8	53.0	56.6	59.5	60.0	57.2	55.1	50.9	49.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	51.3	53.3	57.2	59.3	61.9	66.5	66.9	61.8	59.7	56.6	55.7	(27)
(28)			MCDWB	54.8	53.6	56.6	59.9	64.1	67.3	71.8	71.2	64.9	62.0	57.9	56.6	(28)
(29)		2%	WB	50.4	50.0	51.2	53.6	57.1	59.9	63.4	64.1	60.1	58.3	54.3	52.7	(29)
(30)			MCDWB	52.2	52.6	54.0	57.6	61.2	64.7	68.0	67.8	63.2	60.6	55.9	54.4	(30)
(31)		5%	WB	48.9	48.2	49.7	51.1	55.1	58.4	61.6	62.4	59.0	57.0	52.7	50.8	(31)
(32)			MCDWB	50.6	51.0	52.4	53.8	58.8	62.5	65.6	65.9	61.8	59.3	54.3	52.6	(32)
(33)		10%	WB	47.4	46.3	48.3	49.4	53.6	57.0	59.9	60.5	57.6	55.4	51.1	49.1	(33)
(34)			MCDWB	49.3	48.7	51.0	52.0	57.2	60.8	63.5	63.7	60.3	57.5	52.9	51.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	7.8	8.1	9.0	8.5	8.5	8.9	8.5	7.2	6.6	6.4	(35)
(36)			MCDBR	8.5	9.3	9.4	10.7	12.6	11.8	12.7	11.7	9.9	8.5	8.0	7.2	(36)
(37)			MCWBR	8.3	8.4	7.4	7.8	8.5	7.5	7.9	7.6	7.3	7.1	7.5	6.7	(37)
(38)		5% WB	MCDBR	8.4	9.1	8.9	10.3	11.8	11.4	11.6	11.1	9.7	8.2	7.8	7.4	(38)
(39)			MCWBR	8.5	8.8	7.9	8.1	8.9	8.3	8.2	8.5	8.1	7.2	7.8	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.348	0.372	0.387	0.390	0.384	0.386	0.370	0.364	0.348	0.329	0.326	(40)	
(41)		taud	2.239	2.320	2.299	2.298	2.338	2.390	2.390	2.448	2.455	2.427	2.308	2.204	(41)	
(42)		Ebn at Noon	182	230	252	265	271	274	271	269	254	229	186	156	(42)	
(43)		Edh at Noon	19	25	32	36	37	35	35	31	27	22	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	172	389	749	1213	1551	1561	1474	1207	844	466	221	127	(44)	
(45)		RadStd	19	38	62	96	136	155	126	92	53	50	23	13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page